MIL-S-55498A(EL)
30 January 1970
Superseding
MIL-S-55498(EL)
25 October 1966

MILITARY SPECIFICATION
SHELTER FACILITY,ELECTRICAL EQUIPMENT
S-371()/MSC-31A

1. SCOPE

1.1 Scope.- This specification covers the air or vehicular transportable shelter facility part of the Operation Center, Communications AN/MSC-31A which contains telephone switchboard, teletypewriters,local telephone circuits, and provisions for optional installation of radio equipment for use in area-type communication system. (See 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation,Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)
MIL-W-6858	Welding,Resistance,Aluminum, Magnesium,Non-Hardening Steels or Alloys,Nickel Alloys,Heat-Resisting Alloys and Titanium, Spot and Seam
MIL-T-7928	Terminals,Lug and Splice,Crimp Style Copper
MIL-M-13231	Marking of Electronic Items

FSC 5895

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MIL-F-14072	Finishes for Ground Signal Equipment
MIL-W-45205	Welding,Gas,Metal-Arc & Gas,Tungsten-Arc,Aluminum Alloys,Readily Weldable,for Structures,Excluding Armor
MIL-S-55505	Shelter,Electrical Equipment(Containing Equipment)Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-252	Wired Equipment,Classification of Visual and Mechanical Defects
MIL-STD-1235	Single and Multilevel Continuous Sampling procedures & tables for inspection by attributes.

DRAWINGS

ELECTRONICS COMMAND

SC-DL-543227	Shelter,Electrical Equipment, S-371()/MSC-31A
SC-D-543370	Power Schematic Diagram, S-371()/MSC-31A
SC-D-543365	Signal Schematic Diagram, S-371()/MSC-31A
SC-D-543460	Power Wiring Diagram, S-371()/MSC-31A

(Copies of specifications,standards,drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

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3. REQUIREMENTS

3.1 Description.- Shelter facility,Electrical Equipment S-371()/MSC-31A (hereinafter referred to as shelter facility) is a modified S-280()/G shelter and contains all brackets,mountings, tables, tiedowns,wiring,facilities, and ancillary items which are needed for the installation and operation of the components of the AN/MSC-31A.

3.2 Preproduction sample.- Unless otherwise specified in the invitation for bids and contract, the contractor shall furnish one preproduction sample of the shelter facility for approval. (See 4.3)

3.3 Operational equipment.- Where specified herein the following Government loaned equipment shall be installed in the shelter facility to be tested unless otherwise specified in bid request or contract (6.1b).

2 ea Heater,1500 W,115 VAC
1 ea Switchboard,Telephone Manual SB-22()/PT
3 ea Telephone Set TA-312/PT
1 ea Intercommunication Station LS-147C/F1
1 ea Headset-Microphone H-144C/U
1 ea Air conditioner, 9000 BTU/HR, 115 VAC

3.4 Construction.- The shelter facility shall be constructed in accordance with drawing list SC-DL-543227 (see 4.17).

3.5 Electrical installation.- Shelter facility S-371()/MSC-31A.

3.5.1 Power wiring,shelter facility.- The shelter facility shall be wired in accordance with -

Power Schematic Diagram	SC-D-543370
Power wiring diagram	SC-D-543460

and shall comply with the requirements of this specification.

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3.5.1.1 Operation of distribution panel circuits.- All components on all circuits of the distribution panel shall meet the continuity and operational tests described in 4.6.1.1 through 4.6.1.8.

3.5.1.2 Dielectric breakdown and insulation resistance, power wiring.- All AC power wiring shall have an insulation resistance of not less than 100 megohms when measured between

a. "Hot line" and "Neutral"
b. "Neutral" and "Ground"
c. "Hot line" and "Ground"

Insulation resistance shall be measured at least 10 seconds after an instantaneous potential of 500 volts D.C. is applied (4.6.1.9):

3.5.2 Audio signal wiring,shelter facility.- The shelter facility shall be wired in accordance with drawings listed on SC-DL-543227 and shall comply with the requirements of this specification.

3.5.2.1 Distribution and continuity,audio signal wiring.- There shall be electrical continuity between each terminal as shown as drawing SC-D-543365 (see 4.6.2.1).

3.5.2.2 Patch cords.- Patch cords shall be tested in accordance with 4.6.2.2.

3.5.2.3 Dielectric breakdown and insulation resistance, audio signal wiring.- All audio signal wiring including individual wires,twisted pairs, cables harnesses, and cable assemblies shall have an insulation resistance of not less than 100 megohms when measured between each conductor and the remaining conductors, metallic connector shells, conduit, wireduct,and shelter ground. Insulation resistance shall be measured at least 10 seconds after an instantaneous potential of 500 volts D.C. is applied (see 4.6.2.3).

3.5.3 Cable assemblies,power and audio frequency.- All power, audio frequency cable, and cord assemblies shall be fabricated in accordance with the drawings unless otherwise stated and meet the requirements of MIL-C-3885.

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3.6 Military truck transport.- The shelter facility shall be capable,without any extra tieing,taping,etc., of being transported over cross-country terrain by military vehicle and shall withstand the test of 4.7. After testing per 4.7, there shall be no deformation,buckling,delamination,or damage to any part of the shelter facility.

3.7 Rail impact test.- The shelter facility shall be capable, without any extra tieing, taping,etc, of being loaded, blocked and braced on a flatcar for shipment by rail and shall withstand the test of 4.8. After each impact, there shall be no deformation, buckling,delamination or damage to any part of the shelter facility.

3.8 Drops,flat and rotational.- The shelter facility shall withstand drops of 18 inches onto concrete. After each of the drop tests of 4.9.1 and 4.9.2, there shall be no deformation, buckling,delamination, or damage to any part of the shelter facility.

3.9 Watertightness.- The shelter facility shall withstand the test of 4.10. After the test, there shall be no leakage into the shelter facility or its walls,floor,roof,or doors.

3.10 Fording.- The shelter facility shall withstand the test of 4.11. After the test, there shall be no leakage into the shelter facility or its walls, floor, or doors.

3.11 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive and when installed each insert shall be capable of withstanding the tensile load and torque load referenced in paragraph 4.13. There shall be no damage to the insert threads, the insert shall not have rotated from its original position, and there shall be no distortion of the panel,mounting member,or insert.

3.12 Cleaning.-

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3.12.1 Parts.- After fabrication,parts shall be cleaned in accordance with good commercial practice or as specified in an applicable portion of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly,units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.12.2 Units.- After assembly,the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware, and other foreign material.

3.13 Finish,protective.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings (see 4.4).

3.14 Marking.-

3.14.1 General.- Marking shall conform to Specification MIL-M-13231 and the drawings.

3.14.2 Nameplates.- Shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.14.3 Aircraft loading data plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the centers-of-gravity of the shelter facility and the assemblage in the three reference planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The locations of the centers-of-gravity shall be determined by the contractor using the preproduction sample.

3.14.4 Interior and exterior marking.- Interior and exterior markings shall be as depicted on the drawings.

3.15 Riveting.- Riveted joints shall be tight, the joined points shall be undamaged,and the rivet heads shall be properly seated and tight against their bearing surfaces.

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3.16 Seal,water vapor.- All joints of the framing for the exhaust blowers,power,signal,entrance boxes,and any other framed openings shall be sealed to provide a barrier against the entrance of water vapor to the core material of the shelter panels and to the interior of the shelter.All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets,screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be a one-part sealant,Silicone Construction Sealant SE-1204,as made by General Electric Corp., or equal.

3.17 Soldering.-

3.17.1 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections but no acid or acid salts shall be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.17.2 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux,brush bristles,or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.18 Tropicalization of material.- Canvas and materials as indicated on the drawings shall be treated to be resistant to moisture and fungi as follows:

3.18.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatable with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

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3.18.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished item when used for the purpose intended.

3.18.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements.

3.18.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use (see 6.1).

3.19 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finished weld will be unnecessary except where grinding is specified on the drawings. Inert gas-shielded arc welding (MIG) using helium,argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 CL.B and MIL-W-45205 CL.B as applicable; surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks,cracks,voids,slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4)

3.20 Wiring and cabling.- Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

3.20.1 Cabling.- Insulated wires shall be formed into cables as indicated on drawings. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

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3.20.2 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided for at least two replacements of the part to which the wire or cable is connected.

3.20.3 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.20.4 Splicing.- Wires shall not be spliced.

3.20.5 Connections.-

3.20.5.1 General.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver-plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.20.5.2 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928 (See 4.12).

3.21 Interchangeability.- Like units, assemblies, sub-assemblies, and replaceable parts, shall be physically and functionally interchangeable without modifications of such items or of the equipment (See 4.15). Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on unspecified dimension, rating, characteristic, etc.

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3.22 Workmanship.- The equipment shall be manufactured and assembled in accordance with applicable portions of the following paragraphs:

3.4 Construction
3.5 Electrical Installation
3.11 Mounting inserts
3.12 Cleaning
3.13 Finish
3.14 Marking
3.15 Riveting
3.16 Seal,water vapor
3.17 Soldering
3.18 Tropicalization of material
3.19 Welding
3.20 Wiring and cabling
3.21 Interchangeability

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be as classified as follows:

a. Preproduction inspection (does not include preparation for delivery) (See 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

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(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5)

(2) Quality conformance inspection of preparation for delivery (see 4.18).

4.3 Preproduction inspection.- This inspection shall be performed by the contractor on the preproduction unit unless otherwise specified in the contract. (See 3.2) It shall consist of the inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (see Tables II,III, and IV, respectively). The preproduction inspection shall be performed in this order, unless otherwise specified:

a. Mounting inserts
b. Power wiring
c. Signal wiring
d. Vehicular transportation
e. Rail transportation
f. Flat drop
g. Rotational drops
h. Watertightness
i. Fording

The last tests performed shall be a repeat of the electrical tests. Remaining preproduction tests specified for Groups A,B and C inspections but not included in the above list shall be performed in any order satisfactory to the contracting officer.

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Table I - Preproduction Inspection

Inspection (For additional preproduction inspection see 4.3)	Req Para	Insp Para
Vehicular transportation	3.6	4.7
Rail transportation	3.7	4.8
Mounting inserts	3.11	4.13.1

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where Required
Cable assemblies	3.5.3
Finish	3.13
Marking	3.14
Welding	3.19
Terminals	3.20.5.2

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.5) Test equipment for government verification inspection shall be made available by the contractor.

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4.5.1 Group A inspection.- Each shelter facility shall be inspected for conformance to all the inspection and test requirements of Table II. Equipment so conforming, shall be subject to verification utilizing the procedures for the multilevel continuous sampling plan (CSP-M) of MIL-STD-1235, unless otherwise specified.

Table II Group A Inspection

Inspection	Req Para	Insp Para	AQL% Major	AQL% Minor
Visual and Mechanical	3.22	4.16	1.0	4.0
Electrical				**
Power wiring	3.5.1	4.6.1		
Operation of distribution panel circuits	3.5.1.1	4.6.1.1 through 4.6.1.8	1.0% for the entire group of power and signal wiring	
Dielectric breakdown and insulation resistance				
Power wiring	3.5.1.2	4.6.1.9		
Audio signal wiring	3.5.2	4.6.2		
Distribution and continuity	3.5.2.1	4.6.2.1		
Patch cords	3.5.2.2	4.6.2.2		
Dielectric Breakdown	3.5.2.3	4.6.2.3		

** All electrical defects are considered major

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4.5.1.1 Group A sampling plan.- The Group A sampling plan for the AQL's listed in Table II above when used in conjunction with Table X of standard MIL-STD-1235, shall be as follows:

AQL%	f	K	i	k 1	k 2	k 3
1.0	½	2*	43	½	¼**	Not to be used
4.0	½	3*	15	½	¼	1/8 **

* Max "K" inspection level permitted
** Max "k" sampling freq permitted

4.5.2 Group B inspection.- This inspection, including sampling shall conform to Table III and to the procedures for multi-level continuous sampling plan (CSP-M) of Standard MIL-STD-1235, unless otherwise specified. Group B inspection shall normally be performed on units of production that have passed Group A inspection. The government reserves the right to require complete retest of any sample which is reworked after failing Group B inspection.

Table III - Group B Inspection

Inspection	Rqr Para	Insp Para	AQL% (see 4.5.2.1)
Mounting insert	3.11	4.13.2	4.0
Watertightness	3.9	4.10	2.5
Fording	3.10	4.11	2.5
Interchangeability	3.21	4.15	4.0

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4.5.2.1 Group B sampling plan.- The Group B sampling plan for the AQL listed in Table III, above, when used in connection with Table X of MIL-STD-1235, shall be as follows:

AQL%	f	K	i	k 1	k 2	k 3
2.5	½	2*	20	½	¼**	Not to be used
4.0	½	3*	15	½	¼	1/8**

* Max "K" inspection level permitted
** Max "k" sampling freq permitted

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection.

Table IV - Group C Inspection.-

Inspection	Rqr Para	Insp Para
Construction verification	3.4	4.17
Drops,flat and rotational	3.8	4.9
Solderless-terminal-lug-connections	3.20.5.2	4.14

4.5.3.1 Sampling for inspection of construction evaluation.- The Equipment Verification Review, in accordance with paragraph 4.17, shall be performed once on each part and assembly, randomly selected, and their installation into one of first 10 shelter facilities produced.

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4.5.3.2 Sampling for inspection of drop tests.- One sample shall be selected at random from the first 50 units or fraction thereof, and one sample from each additional 150 units or fraction thereof, produced and subjected to the drop tests.

4.5.3.3 Sampling for inspection of solderless-terminal-lug-connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1 foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.4 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation of details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of production which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed; at the option of the government) and Groups A and B inspection may be reinstituted; final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. (See 6.3)

4.5.4 Reinspection of conforming Group B and Group C Sample Units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage. (See 6.3)

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4.6 Electrical installation tests.- The tests of 4.6.1 and 4.6.2 shall be performed on each shelter facility. Before applying power to the shelter facility, an electrical ground at the power entrance box shall be made. Power to be supplied shall be 115 VAC, 60 cycles, single phase.

4.6.1 Power wiring, facility.- The power wiring installation shall be inspected for conformance with drawings referenced in 3.5.1.

4.6.1.1 Test of main breaker circuit.-

a. Remove neon power indicator lamp. Switch main breaker to off.

b. Using a low voltage indication device (such as a buzzer) verify continuity between terminals 1 of one Power Receptacle U-238()/G and terminal 1 of other U-238()/G in Power entrance box.

c. Repeat b for terminals 2 and 4 of both receptacles.

d. Install neon power indicator lamp.

e. Connect a 115 VAC, 60 cycle power source to the facility through one of the U-238()/G power receptacles. Neon power indicator lamp shall be lighted.

f. Switch on main breaker. Neon power indicator lamp shall remain lighted. Voltmeter shall indicate nominal 115 VAC.

4.6.1.2 Test of breaker No. 1 circuit.-

a. Switch on breaker No.1, neon indicator shall light.

b. Turn on blackout bypass switch. Incandescent lamp shall light.

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c. Turn on light switch No.1. Alternate fluorescent lamps shall light.

d. Turn on light switch No.2. Remaining fluorescent lamps shall light.

e. Close door and turn off blackout bypass switch. All lamps shall remain lighted.

f. Unlatch door. All lamps shall extinguish before latch is in a full open position. Neon power indicator lamp shall remain lighted.

g. Relatch door. All lamps shall be lighted when latch is in the locked position.

h. Repeat f and g twiece.

i. Turn on blackout bypass switch and leave on for remainder of inspection. All lamps shall be lighted.

j. Switch breaker No.1 off and then on. All lamps shall alternately extinguish and relight.

4.6.1.3 Test of breaker No.2 circuit.-

a. Switch on breaker No.2, neon indicator shall light. (Check that vent cover is open). Plug in blower No.1.

b. Turn on switch for blower No.1. Blower No.1 shall operate and shall exhaust air from shelter.

c. Turn off switch for blower No.1. Blower shall stop.

d. Plug 115 VAC test lamp into blower No.2 receptacle. Turn on switch for blower No.2. Test lamp shall light.

e. Turn off switch for blower No.2. Test lamp shall extinguish.

f. Turn on switches for both blower No.1 and No.2. Switch off breaker No.2. Blower shall stop and test lamp shall extinguish. Remove test lamp.

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4.6.1.4 Test of breaker No.3 circuit.-

a. Switch on breaker No.3, neon indicator shall light.

b. Plug a 115 VAC test lamp into Heater No.1 receptacle. Test lamp shall light in the receptacle.

c. Switch off breaker No.3. Test lamp shall extinguish at the receptacle.

4.6.1.5 Test of breaker No.4 circuit.- Repeat test in paragraph 4.6.1.4 for breaker No.4 and heater No.2.

4.6.1.6 Test of breaker No.5 circuit.-

a. Switch on breaker No.5, neon indicator shall light.

b. Plug 115 VAC test lamp into each side of three "Convenience" duplex receptacles. Test lamp shall light in each position.

c. Switch off breaker No.5. The test lamp shall not light when plugged into any of the "convenience" receptacles.

4.6.1.7 Test of breaker No.6 circuit.-

a. Switch on breaker No.6, neon indicator shall light.

b. Plug 115 VAC test lamp into each equipment receptacle. (Intercom, AN/GRR-5, PP-2953/U). Test lamp shall light in each position.

c. Switch off breaker No.6. The test lamp shall not light when plugged into any equipment receptacle.

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4.6.1.8 Test of breaker No.7 circuit.-

a. Switch on breaker No.7, neon indicator shall light.

b. Plug a 115 VAC test lamp into air conditioner power receptacle on exterior of shelter. Test lamp shall light.

c. Switch off breaker No.7. Test lamp shall extinguish.

4.6.1.9 Dielectric breakdown and insulation resistance power wiring.- The dielectric breakdown and insulation resistance of the power wiring shall be tested for conformance with 3.5.1.2 as described below.

a. Remove all fluorescent lamps from their fixtures.

b. Remove all starters from fluorescent fixtures.

c. Remove cover from power distribution box. Disconnect voltmeter from circuit. Disconnect all green (ground) wires from neutral bus bar in power distribution box. Switch on all circuit breakers and switches, except circuit breaker for lights.

d. Remove all incandescent and neon lamps.

e. Unplug all power cords (blowers,air conditioner, heaters, intercom, radio, power supply etc) from receptacles in wire-duct.

f. Apply an instantaneous voltage of 500 volts D.C. between terminal 1 of Power Receptacle U-238()/G and terminal 2 of the same receptacle. After 10 seconds measure the resistance between terminal 1 and terminal 2. Resistance shall not be less than 100 megohms.

CAUTION.- After each test, discharge the conductor under test to ground to eliminate danger of shock.

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g. Repeat f replacing terminal 2 (2 places) with terminal 4.

h. Repeat f replacing terminal 1 (2 places) with terminal 4.

i. Restore shelter facility to original condition; that is, replace all lamps, cover, wires, etc.

4.6.2 Audio signal wiring, facility.- The audio signal wiring installation shall be inspected for conformance with the drawings referenced in 3.5.2. Jack designation (such as J3, etc) referenced in following tests are identified on signal schematic diagram SC-D-543365.

4.6.2.1 Distribution and continuity, audio signal wiring.- The distribution and continuity of the audio signal wiring shall be tested for conformance with 3.5.2.1 by using a low voltage indication device (hereafter referred to as continuity tester) such as a buzzer as described below:

a. Remove all plugs (telephone and intercom) from jacks in wireduct. Disconnect all audio lines (1A thru 12B) from the SB-22()/PT and check that the lines are not shorted to each other, the shelter, or any other conducting material.

b. Connect one lead of the continuity tester to the male contact 1A in U-187A/G (signal 1) and the other lead to binding post 1A. Continuity shall be indicated (i.e. buzzer shall sound). Repeat this test for each male contact in U-187A/G and the corresponding binding post up to and including 26B (for a total of 52 tests). Continuity shall be indicated in each case.

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c. Connect one lead of the continuity tester to the binding post 1A and the other lead to the wire end (for SB-22()/PT) of Wht-Blu wire. Continuity shall be indicated. Repeat this test for each binding post and wire end up to and including 12B and Wht-Org-Red wire (for a total of 24 tests). Continuity shall be indicated in each case.

d. Connect one lead of the continuity tester to the binding post 22A and the other lead to the tip connection of a dummy plug (PG-047B) inserted into jack J3. Continuity shall be indicated.

e. Repeat d. for each of the following combinations (for a total of 12 tests):

Binding post 22B and sleeve of jack J3

Binding post 23A and tip of jack J5

Binding post 23B and sleeve of jack J5

Binding post 24A and tip of jack J7

Binding post 24B and sleeve of jack J7

Binding post 26A and tip of jack J4

Binding post 26B and sleeve of jack J4

Binding post 26A and tip of jack J6

Binding post 26A and tip of jack J2

Binding post 26B and sleeve of jack J2

4.6.2.2 Patch cords.- Using the continuity tester, test between the tips of each plug on each patch cord, between the sleeves of each plug on each patch cord, and between one sleeve and one tip of each patch cord. Continuity shall be indicated in each of the first two tests, and discontinuity shall be indicated in the third test. (3.5.2.2)

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4.6.2.3 Dielectric breakdown and insulation resistance, audio signal wiring.- The dielectric breakdown and insulation resistance of the audio signal wiring shall be tested for conformance with 3.5.2.3 (with either manual or automatic testing equipment) as described below:

a. Remove all telephone and intercom plugs from jacks in wireduct and patch panel. Check that audio lines 1A thru 12B are disconnected from SB-22()/PT and are not shorted to each other, the shelter, or any other metallic object.

b. Connect (51) binding posts 1B through 26B directly to shelter ground (in Power entrance box). Apply an instantaneous voltage of 500 volts DC between binding post 1A and shelter ground. After 10 seconds measure the resistance between binding post 1A and shelter ground. Resistance shall be not less than 100 megohms.

CAUTION: After each test, discharge the conductor under test to ground to eliminate danger of shock.

c. Repeat b for each of the remaining 51 audio signal lines; that is, the 500 v potential shall be applied in turn between binding posts 1B through 26B and shelter ground while the remaining 51 binding posts are connected directly to shelter ground, (for a total of 52 tests).

4.7 Military truck transport.- The shelter facility with operational equipment (3.3) installed shall be transported on a Government furnished M-35 truck over the specially prepared course described in paragraph 4.7.1 for 10 laps at 5 mph, 10 laps at 10 mph, 10 laps at 15 mph, and 10 laps at 20 mph. A lap is defined as traversing the course in one direction. The shelter facility shall meet the requirements of 3.6 after completion of the above test.

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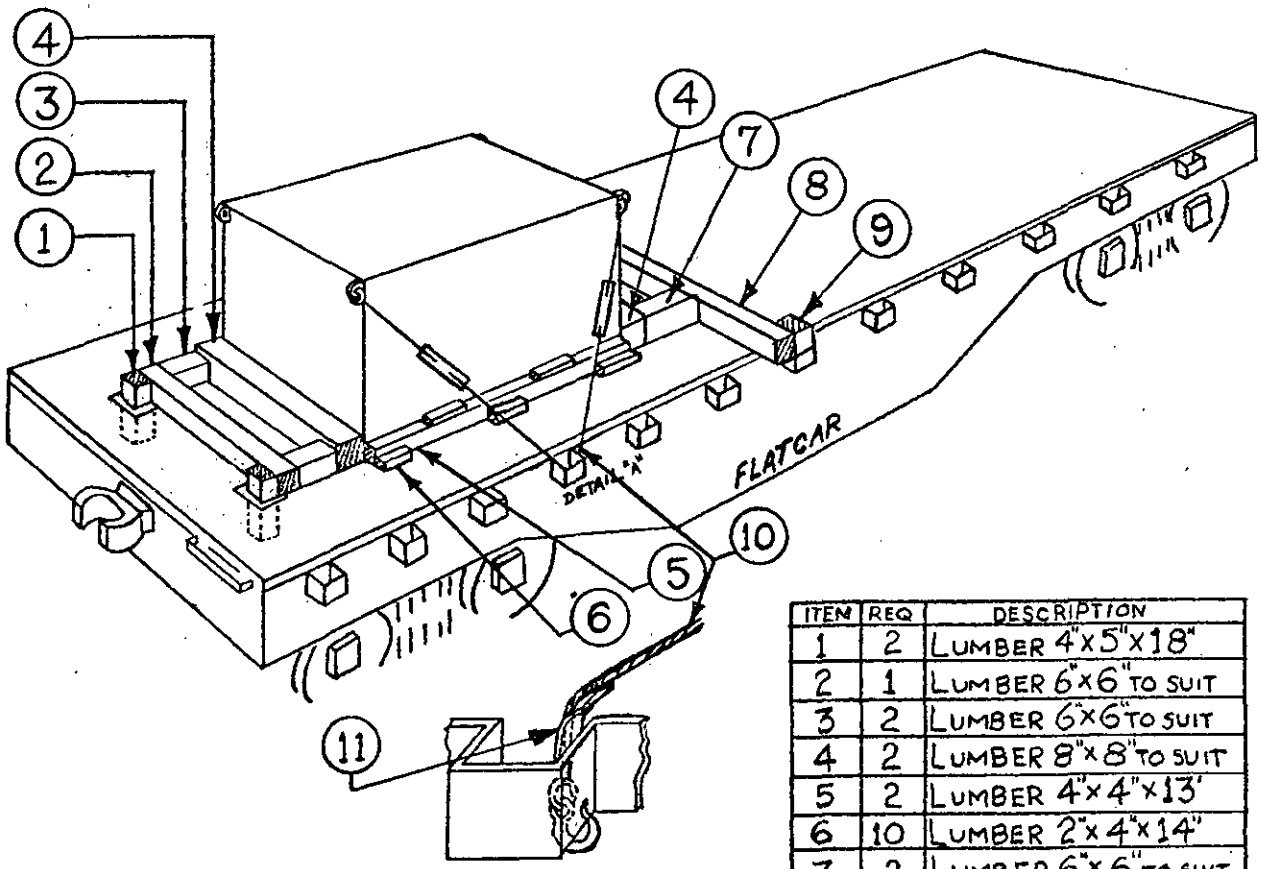
4.7.1 Military truck transport course.- The course for the military truck transport test shall have twelve 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above the ground. The edges shall have a 1-inch by 1-inch bevel. The sixth and twelfth timbers shall be placed 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be long enough to span the vehicle passing over them and shall be anchored securely by any convenient means. Ground on which timbers are placed shall be hard and level.

4.8 Rail transport, impact test.- The shelter facility with operational equipment (3.3) installed shall be loaded on a railroad flatcar (test car) in accordance with Figure 1. Lashing shall consist of 4 wire rope slings each with minimum breaking strength of 9000 pounds and each including 3/8-inch diameter, 6 x 19, flexible, steel, preformed-wire rope (having a breaking strength of 10,600 pounds minimum), a turnbuckle, two hooks, with thimbles and fasteners as required. Before each impact, the slings shall be taut (turnbuckle shall be hand-tight plus no turn). The test shall be conducted on a flat stretch of track. A loaded car totaling 165,000 pounds (impact car) and traveling a 9 mph (or equivalent impact energy conditions approved by the Government) shall be impacted against the test car coupled to two other empty cars. These cars shall be stationary with all brakes off. Four impacts shall be performed, two impacts into each end of the test car. The cabling and blocking holding the shelter facility in place shall be tight at the start of each impact run. If the cable or blocking is torn loose or if the load on the impact car shifts, the run shall be repeated. A damage survey shall be made after each impact and the shelter facility shall meet the requirements of 3.7.

4.9 Drop tests.-

4.9.1 Flat drop.- The shelter facility with operational equipment (3.3) installed shall be lifted and allowed to fall freely a distance of 18 inches with the skids impacting onto a hard concrete surface. This test shall be performed once. Upon completion of this drop, the shelter facility shall meet the requirements of 3.8.

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DETAIL "A"

Wire tiedown same for all points of contact with stake pockets

ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"X5"X18"
2	1	LUMBER 6"X6" TO SUIT
3	2	LUMBER 6"X6" TO SUIT
4	2	LUMBER 8"X8" TO SUIT
5	2	LUMBER 4"X4"X13'
6	10	LUMBER 2"X4"X14"
7	2	LUMBER 6"X6" TO SUIT
8	1	LUMBER 6"X6" TO SUIT
9	2	LUMBER TO SUIT
10	16	SLING (SEE PARA 4.8)
11	4	FLEX. METAL SHEETING

NOTES:

a. The blocking lumber shall be notched to accomodate towing eyes, bolts and rivet heads.

b. The blocking lumber shall be rigidly fastened to the floor of the flatcar.

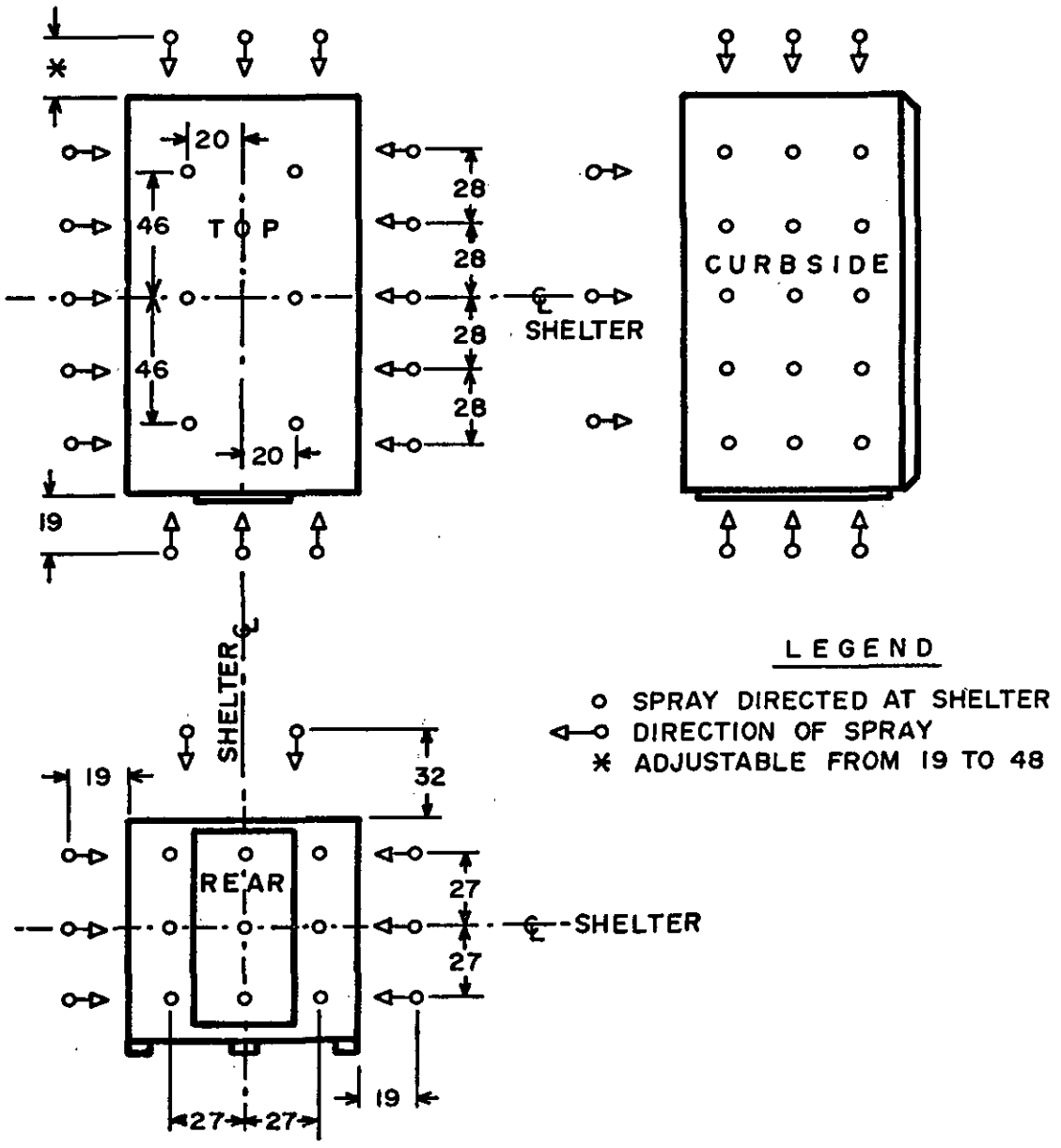
FIGURE #1

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4.9.2 Rotational drop.- The shelter facility with operational equipment (3.3) installed shall have one edge placed on a nominal 4-inch high board. The opposite edge of the shelter facility shall be raised 18 inches and allowed to fall freely onto a hard concrete surface. This drop test shall be performed on each bottom edge of the shelter facility. Upon completion of these drops, the shelter facility shall meet the requirements of 3.8. In each drop test, the 4-inch high member shall be placed under the skids, and the eighteen inch measurement shall be taken from the bottom of the skids.

4.10 Watertightness test.- The shelter facility without the use of any additional caulking, sealing, taping, etc, shall be subjected to the following test (with air conditioner and all associated bracketry removed and air conditioner blank-off panel installed and with inside covers of entrance boxes removed): Each of the five exposed surfaces of the shelter, with covers opened, shall be sprayed with water from nozzles, model G29SQ (or GG29SQ) as made by Spraying Systems Co. Bellwood,Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. (See Figure 2 for nozzle location.) Each panel shall be sprayed for a period of twenty minutes. (If the watertight test is to be followed by the fording test of 4.11, the bottom row of nozzles, with the exception of the nozzles facing the door-end panel, can be eliminated.) Upon completion of the above test, the shelter facility shall meet the requirements of 3.9. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the Government at time of testing. After this determination, the holes shall be sealed with rivets and sealer.

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NOTES

1 NOZZLES SHALL BE PERPENDICULAR TO SHELTER.
2 UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3 GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4 ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5 FRONT NOZZLE LOCATIONS SAME AS REAR.
6 DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE 2

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4.11 Fording test.- The shelter facility, without the use of any additional sealer, caulking, taping, fording kit, etc, shall be immersed in water to a depth of twenty-one inches, measured from the bottom of the shelter skids, for a period of one hour. The shelter shall be restrained by external tiedowns. Upon completion of this test, the shelter facility shall meet the requirements of 3.10 To determine that no water has entered space between the skins, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.12 Testing of solderless terminal - lug connections.- Connections made with solderless terminal-lugs shall be tested to meet the requirements of 3.20.5.2.

4.13 Mounting insert tests.-

4.13.1 Preproduction insert test.- Four of each size insert shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table VII. After test, the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

4.13.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample shelter facility. Each selected insert shall be tested in accordance with Table VII. After test, the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

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Table VII - Mounting Inserts Loads

Insert size	Preproduction Test		Production Test	
	Axial(lbs)	Torque (in-lbs)	Axial(lbs)	Torque(in.lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60
#10-32	600	23	300	23

4.14 Testing of solderless-terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.20.5.2.

4.15 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

b. Size and mounting dimensions of racks, patch panels and cabinets.

4.16 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.22 and Table VIII.

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Table VIII - Classification of Visual & Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or mislocated
	2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible.
	3. Welds - cracked or porous.
	4. Sealer improperly applied (holes, separations, or lack of adhesion).
	5. Mounting inserts missing or mislocated.
	6. Burrs or sharp edges in wire duct not removed.
	7. Equipments or bracketry loose or missing.
	8. Doors,etc inoperative.
	9. Electrical receptacles, switches, connectors,wire ducts or other electrical components loose,improperly located or inoperative.
	10. Locking or holding devices missing or inoperative.
	11. Gaps between sections or wire duct exceed 1/16"
Minor	1. Doors,not easily operable.
	2. Fastening devices difficult to operate.
	3. Finish - abrasions or scratches.
	4. Burrs & sharp edges on brackets not removed
	5. Screws loose.

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4.17 Construction verification.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings (see 3.4) cited in the Technical Data Package (as supplemented by all approved engineering changes). EVR shall consist of the following:

a. An audit to establish that "as-built" shelter facilities, including individual parts, their use in subassemblies and assemblies (components), and their final installation into the shelter forming the shelter facility, are in accordance with the end product drawings.

b. An audit to establish that "as-built" shelter facilities,including parts, components and their installation, meet the acceptance requirements specified in the Technical Data Package for each shelter part, component, and shelter facility. The audit shall record all acceptance test methods used together with resulting verification test data.

c. Preparation of EVR report(s) containing the following information:

(1) Identification of parts,components,shelter facilities and details proving compliance with 4.17.a and b above.

(2) Discrepancies noted.

(3) Corrective action taken.

d. Two (2) copies of the EVR report shall be submitted no later than 15 days prior to the submission of the first production unit or lot to the government technical organization cited in the contract. The results of the EVR shall be subject to verification by authorized government personnel at the time the first unit or production lot is offered for acceptance. Government verification will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to government personnel during the verification. The contractor shall provide the following:

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(1) Segregation of parts, components of shelter facilities, to permit reviewing government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the shelter facility necessary to permit government review of the shelter facility, its components and their installation, and to reassemble parts after inspection.

(2) Have as required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

4.18 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.19 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Paragraph 4.6.1 and 4.6.2	Power and Signal wiring
Paragraph 4.16	Visual and mechanical

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation,packaging, and packing of the shelter facility shall be in accordance with applicable requirements of Specification MIL-S-55505.

6. NOTES

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6.1 Ordering data.- Procurement documents should specify the following:

a. Title,number, and date of this specification and any amendment thereto.

b. List of each individual equipment item to be used in testing the shelter facility (3.3).

c. Submission of the statement of treatment referenced in 3.18.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.2 Definitions.-

6.2.1 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.2 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.2.3 Testing.- Testing consists of determinations of compliance, using technical means.

6.3 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government,pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.4).

6.4 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: S-371 X/MSC-31A. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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6.5 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:

Army - EL

Preparing Activity:

Army - EL

Project No. 5895-A099

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-S-55498A(EL), SHELTER FACILITY, ELECTRICAL EQUIPMENT S-371( )/MSC-31A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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